

Train Schedule Analysis and Interactive Route Enquiry System Using Python

INTERNSHIP PROJECT

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Project Introduction

- Analysis of train schedule data using Python
- Data processing, quality checking, analysis and visualization
- Interactive route enquiry for direct train information
- Developed as an internship project prototype
- Users can enter source and destination stations to find direct trains.

Project Objectives

- Analyze train schedule data
- Identify train routes, stations and number of stops
- Calculate journey duration
- Classify routes and analyze station frequency
- Perform data quality checks
- Create visualizations and an interactive route enquiry system

Technologies Used

- Python
- Pandas – data processing and analysis
- NumPy – numerical operations
- Matplotlib – data visualization
- Jupyter Notebook – development environment
- CSV dataset

Level 1 – Basic Data Review

- Dataset overview and structure
- List trains with starting and ending stations
- Calculate stops for each train
- Identify trains with maximum and minimum stops
- Understand the basic structure of the schedule data





Level 2 – Simple Data Processing

- Standardize arrival and departure times
- Calculate total journey duration
- Classify routes into Short, Medium and Long
- Analyze station-wise train frequency
- Prepare processed data for further analysis

LEVEL 3 & 4 – DATA QUALITY AND ANALYSIS

- DATA QUALITY CHECKS:
- MISSING-VALUE CHECKING
- DUPLICATE-RECORD CHECKING
- STATION-ORDER VALIDATION
- ANALYSIS & VISUALIZATION:
- AVERAGE JOURNEY DURATION
- HIGH-TRAFFIC STATIONS AND BASIC CHARTS

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LEVEL 5 – ADVANCED ANALYSIS & VISUALIZATION

- PIVOT TABLES FOR STATION AND TRAIN ANALYSIS
- CROSS-TABULATION OF ROUTE TYPES
- COMPARATIVE CHARTS
- NUMBER OF TRAINS BY ROUTE TYPE
- AVERAGE JOURNEY DURATION BY ROUTE TYPE
- ADVANCED OBSERVATIONS FROM THE DATASET

LEVEL 6 – INTERACTIVE ROUTE ENQUIRY

- USER ENTERS SOURCE STATION AND DESTINATION STATION
- SYSTEM SEARCHES FOR DIRECT TRAINS
- OUTPUT INCLUDES:
- TRAIN NUMBER
- SOURCE AND DEPARTURE TIME
- DESTINATION AND ARRIVAL TIME
- ESTIMATED JOURNEY DURATION

Conclusion & Future Scope

- Completed structured train schedule analysis using Python
- Performed data processing, quality checks and visualization
- Developed an interactive route enquiry prototype

Future scope: web/mobile interface, live train information, larger database and real-time deployment

The End